UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

UTM Johor Bahru

SECV 2223

WEB PROGRAMMING

Semester 02, 2025/2026

Assignment 1: Report

Lecturer:

Ts. Dr. Sarina binti Sulaiman

Representative:

Group Members:

Prepared by:

MOHAMAD ASYIK ILLAHI BIN NASURIDDIN	A24CS0113
NORHASLIN BINTI SHARI	A24CS0149
FATIN SYAKIRAH BINTI MUHAMAD HARDON	A24CS0073
BATRISYIA AMANI BINTI KHAIRUN HALED	A24CS0054

Table of Content

---

1. Abstract (17 April 2026)

This report evaluates eight leading e-commerce platforms (Zalora, Lazada, Mudah, SHEIN, TikTok Shop, Carousell, BlendJet, and HiSmile), using Google Lighthouse to assess their Performance, Accessibility, Best Practices, and Search Engine Optimization (SEO). The findings indicate that while major marketplaces like Lazada possess high SEO authority, they often struggle with technical performance due to complex interfaces and heavy third-party scripts. In contrast, platforms such as Zalora and BlendJet demonstrate superior accessibility and faster loading speeds. By identifying these specific technical strengths and weaknesses, the report offers data-driven recommendations to improve market competitiveness in the digital retail economy.

2. Analysis

We will begin our analysis with Zalora. This web shows a high level of technical proficiency in search engine optimization (SEO), achieving a perfect score of 100. The site got a 95 for accessibility, which shows it’s very user-friendly. Despite these strengths, it has the performance rating of 79 and adversely affects loading velocities. Additionally, the Best Practices score of 54 is due to reliance on deprecated APIs and the presence of minor security warnings.

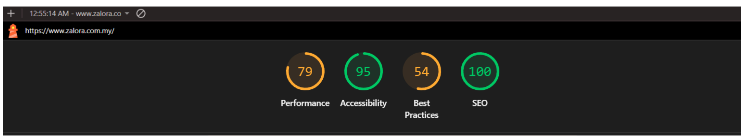

Figure 1: Zalora’s Lighthouse Analysis

Subsequently, Lazada also maintains the performance score of 70, supported by a significant SEO score of 92 that provides a stable visual experience during the loading process. However, the analysis identifies critical deficiencies in accessibility, where the platform scores only 53 due to missing labels, improper ARIA usage, and inadequate color contrast. Technical standards are further compromised by a Best Practices score of 58, resulting from insecure requests and substandard coding practices.

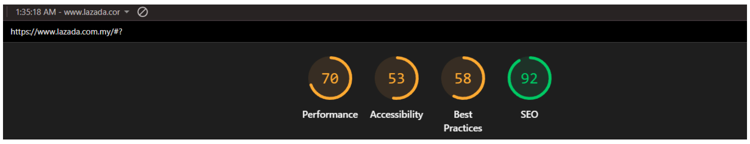

Figure 2: Lazada's Lighthouse Analysis

Next, for the SHEIN website. The results show several domain strengths such as in terms of performance , SEO and accessibility . Performance got a score of 63 indicating fair but very fast rendering times. Besides, SEO scores 92 out of 100, showing a strong optimization and ensuring good visibility in search engines. The accessibility got a perfect score of 100. This shows the website has great labelling and sufficient colour contrast. However, there are few weaknesses for the SHEIN website. Best practice score of only 54 makes it poor due to insecure requests and weak coding standards.

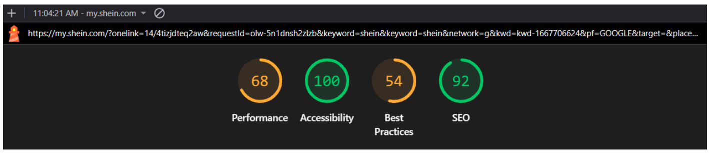

Figure 3: SHEIN's Lighthouse Analysis

<u>18 April 2026</u>

Furthermore, for Tiktok Shop, SEO got a perfect score of 100, showing excellent search engine optimization. The accessibility is also high, which is 83 indicates it has a strong support for users with assistive technology and has fewer issues. In addition, the weakness of this website lies mainly in the performance which has fallen into the "fair" category with a score of 58 and needs to improve in speed and its efficiency. Besides, the best practice score was only 77, affected by reliance on deprecated APIs and minor security warnings.

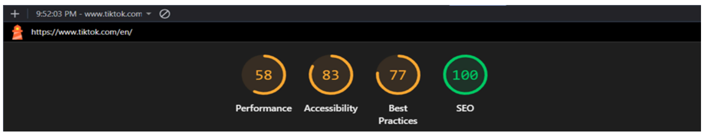

Figure 4: TikTok Shop's Lighthouse Analysis

Then, Mudah website demonstrates a strong foundation in user experience and visibility, boasting an impressive accessibility score of 95 and a robust SEO ranking of 92. These metrics indicate a well-structured, inclusive site that is easily discoverable. However, significant technical bottlenecks hinder its overall efficiency. The performance score sits at a mediocre 55 due to high Total Blocking Time that creates a noticeable lag in interactivity. Furthermore, the best practices score is hampered by a low 50 due to persistent browser errors. Essentially, while the site appears quickly, its heavy background processes and large file size prevent it from being truly seamless.

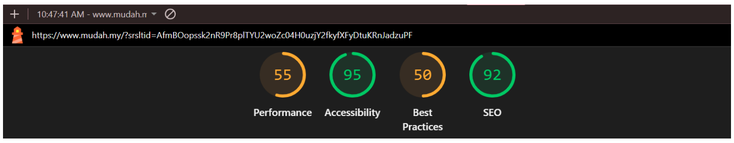

Figure 5: Mudah's Lighthouse Analysis

BlendJet shows several strengths and weaknesses based on its performance results. The website achieved an SEO score of 100, indicating well-optimized structured data. The Total Blocking Time of 150 milliseconds indicates good responsiveness, while a Cumulative Layout Shift of 0 shows stable page elements that reduce misclicks. The Accessibility score of 76 indicates moderate accessibility support. However, weaknesses include a Best Practices score of 54, which highlights the use of deprecated APIs, five third-party cookies, console errors, and some buttons or links lacking accessible names, potentially affecting usability and accessibility.

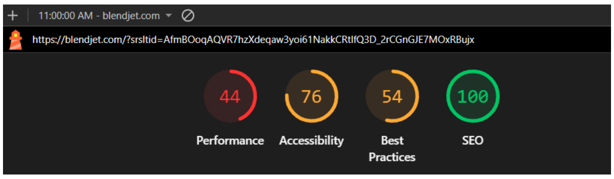

Figure 6: BlendJet's Lighthouse Analysis

3. Comparative Findings (19 April 2026)

After comparing results across all audited websites, there are several clear trends that emerge. First of all, the overall trend that has the strongest area across platforms is Search Engine Optimization (SEO). Most sites score higher than 90 meanwhile Carousell scores only 62 and HiSmile 85. Next, for accessibility, Zalora, Mudah and Carousell show the highest compliance and have the strongest performers. However, Lazada and HiSmile are the weakest among all websites.

Based on the Google Lighthouse performance results, websites show noticeable differences in performance efficiency. Overall, e-commerce platforms like Lazada, SHEIN, and TikTok tend to have moderate performance scores due to heavy images, scripts, and third-party content, which increase loading time. Simpler platforms such as Mudah.my generally perform faster because of lighter page design, while brand-focused sites like Zalora and BlendJet usually achieve balanced performance with faster First Contentful Paint but moderate Total Blocking Time. These differences show that website complexity and third-party resources strongly affect Lighthouse performance metrics such as loading speed, responsiveness, and layout stability.

Among all the websites, the Best Practices scores for all are generally low, showing that many of them still have technical issues. TikTok Shop has the highest score of 77, which means it follows web standards better than the others, although it still has some minor problems like outdated APIs and small security warnings. In contrast, Mudah has the lowest score of 50 due to browser errors and technical issues that affect its performance. The rest of the websites, including Zalora, Lazada, SHEIN, and BlendJet, all have scores between 54 and

58, which are considered poor. These websites commonly face issues such as insecure requests, outdated coding practices, and system errors. Overall, this comparison shows that most of the websites need improvement in following proper web standards, security, and coding practices.

4. Recommendations (20 April 2026)

As for recommendations, Zalora should improve its technical standing by prioritizing the compression of images and the reduction of its overall payload size to facilitate more efficient content loading. Furthermore, the platform is encouraged to maintain its current high standards for search engine optimization and accessibility while addressing technical debt related to the use of deprecated APIs. By resolving these minor security warnings and performance bottlenecks, Zalora can achieve a more seamless and competitive user experience.

Subsequently, Lazada should focus its optimization efforts on enhancing accessibility by incorporating necessary labels, correcting ARIA roles, and improving color contrast for better readability. Additionally, it is recommended that the platform address technical issues involving insecure requests and substandard coding practices. Streamlining payloads and improving caching mechanisms are also essential steps to increase the website's responsiveness and overall performance.

Next, the Shein website needs to focus on making improvements by resolving best practice issues, including insecure requests and weak coding standards. Fixing the streamline payloads and improving caching can help the website to get faster responsiveness. As for Tiktok shop, it is recommended by maintaining strong SEO and accessibility practices so that it can enhance the usability for users. The website also needs to compress images and reduce the payload size so the loading of content can be faster.

Moreover, to further optimize the Mudah site, it is essential to implement efficient caching strategies by setting longer lifetimes for static assets, which minimizes redundant downloads and speeds up return visits. Performance can be significantly enhanced by auditing third-party scripts and removing unnecessary trackers that often bloat the page and delay interactivity. Additionally, focusing on visual stability by preventing unexpected layout shifts will create a smoother user experience, while refining the color palette for better contrast will ensure the content is more readable and accessible for everyone.

To improve BlendJet performance,using tools such as Partytown can be used to offload third-party scripts into web workers and apply server-side tagging, reducing the processing load on users' devices. Static images should be served in AVIF format with responsive srcset attributes, and adopting a headless commerce architecture using frameworks like Next.js or Remix can further improve speed and overall website performance.